Midterm review

Chapter 1 The Foundations: Logic and Proofs

1.1 Propositional Logic Summary

The Language of Propositions

- Connectives: Negation $\neg$ Conjunction $\wedge$ Disjunction $\vee$ Implication $\rightarrow$ Biconditional $\leftrightarrow$ Exclusive Or $\oplus$

- Truth Values
- Truth Tables

Logical Equivalences

- Equivalent Propositions

1.3 Propositional Equivalences Summary

2, Logically Equivalent

Two compound propositions p and q are logically equivalent if $p \leftrightarrow q$ is a tautology.

Two compound propositions p and q are equivalent if and only if the columns in a truth table giving their truth values agree.

We can show that two expressions are logically equivalent by developing a series of logically equivalent statements.

conditional-disjunction equivalence: $\neg p \vee q \equiv p \rightarrow q$

De Morgan's Laws $\neg(p \wedge q) \equiv \neg p \vee \neg q$

$$\neg(p \vee q) \equiv \neg p \wedge \neg q$$

3, Key Logical Equivalences

1.4 Predicate and quantifier summary[1]

Predicates P

Propositional function *P(x)*

Quantifiers

- Universal Quantifier ∀
- Existential Quantifier ∃

Negating Quantifiers

- De Morgan's Laws for Quantifiers

$$\neg\forall x P(x) \equiv \exists x \neg P(x)$$

$$\neg\exists x P(x) \equiv \forall x \neg P(x)$$

1.6 Rules of inference

Inference Rules for Propositional Logic

Rules of Inference for Quantified Statements

Using Rules of Inference to Build Arguments

TABLE 1 **Rules of Inference.**

Rule of Inference	*Tautology*	*Name*
p $p \to q$ $\therefore q$	$(p \land (p \to q)) \to q$	Modus ponens
$\neg q$ $p \to q$ $\therefore \neg p$	$(\neg q \land (p \to q)) \to \neg p$	Modus tollens
$p \to q$ $q \to r$ $\therefore p \to r$	$((p \to q) \land (q \to r)) \to (p \to r)$	Hypothetical syllogism
$p \lor q$ $\neg p$ $\therefore q$	$((p \lor q) \land \neg p) \to q$	Disjunctive syllogism
p $\therefore p \lor q$	$p \to (p \lor q)$	Addition
$p \land q$ $\therefore p$	$(p \land q) \to p$	Simplification
p q $\therefore p \land q$	$((p) \land (q)) \to (p \land q)$	Conjunction
$p \lor q$ $\neg p \lor r$ $\therefore q \lor r$	$((p \lor q) \land (\neg p \lor r)) \to (q \lor r)$	Resolution

TABLE 2 Rules of Inference for Quantified Statements.

Rule of Inference	*Name*
$\forall x P(x)$ $\therefore P(c)$	Universal instantiation
$P(c)$ for an arbitrary c $\therefore \forall x P(x)$	Universal generalization
$\exists x P(x)$ $\therefore P(c)$ for some element c	Existential instantiation
$P(c)$ for some element c $\therefore \exists x P(x)$	Existential generalization

Basic Structures: Sets, Functions, Sequences, Sums, and Matrices

Chapter 2

2.1 Sets

Describing Sets

- Roster Method $S = \{a,b,c,d\}$
- Set-Builder Notation $S = \{x \mid x$ is a positive integer less than 100$\}$

Cardinality of Sets: is the number of (distinct) elements of A, $|A|$.

Power Set: If $A = \{a,b\}$ then $P(A) = \{\emptyset, \{a\},\{b\},\{a,b\}\}$

Tuples: the *ordered n-tuple* (有序n元组)$(a_1,a_2,\ldots..,a_n)$

Cartesian Product

The *Cartesian Product* (笛卡尔积) of two sets A and B, $A \times B$ is the set of ordered pairs (a,b) where $a \in A$ and $b \in B$.

$$A \times B = \{(a,b) \mid a \in A \wedge b \in B\}$$

2.2 Set operations

Set Operations

- Union $\{x \mid x \in A \vee x \in B\}$
- Intersection $\{x|x \in A \wedge x \in B\}$
- Complementation $\bar{A} = \{x|x \in U \wedge x \notin A\}$
- Difference $A - B = \{x \mid x \in A \wedge x \notin B\} = A \cap \overline{B}$

More on Set Cardinality $|A \cup B| = |A| + |B| - |A \cap B|$

Set Identities

Proving Set Identities

Membership Tables

Proving Set Identities

Different ways to prove set identities:

1. Prove that each set (side of the identity) is a subset of the other.
2. Use set builder notation and propositional logic.
3. Membership Tables : Verify that elements in the same combination of sets always either belong or do not belong to the both side of the identity. Use 1 to indicate it is in the set and a 0 to indicate that it is not

2.3 Functions

Floor function, Ceiling function

2.4 Sequences and summations

Sequences.

- Geometric Progression, Arithmetic Progression

Recurrence Relations

- Example: Fibonacci Sequence $f_0 = 0, f_1 = 1, f_n = f_{n-1} + f_{n-2}$

Summations

Sums of terms of geometric progressions

$$\sum_{j=0}^{n} ar^j = \begin{cases} \dfrac{ar^{n+1} - a}{r-1} & r \neq 1 \\ (n+1)a & r = 1 \end{cases}$$

2.6 Matrices

$$A^0 = I_n \qquad A^r = \underbrace{AAA\cdots A}_{\text{r times}}$$

Zero-One matrices

Boolean operations

$$b_1 \wedge b_2 = \begin{cases} 1 & \text{if } b_1 = b_2 = 1 \\ 0 & \text{otherwise} \end{cases} \qquad b_1 \vee b_2 = \begin{cases} 1 & \text{if } b_1 = 1 \text{ or } b_2 = 1 \\ 0 & \text{otherwise} \end{cases}$$

The *join(并)* of **A** and **B** is the zero-one matrix with (i,j)th entry $a_{ij} \vee b_{ij}$. **A** $\vee$ **B.** The *meet(交)* of **A** and **B** is the zero-one matrix with (i,j)th entry $a_{ij} \wedge b_{ij}$. **A** $\wedge$ **B**.

Boolean product **A** $\odot$ **B**, is the $m \times n$ zero-one matrix with(i,j)th entry $c_{ij} = (a_{i1} \wedge b_{1j}) \vee (a_{i2} \wedge b_{2j}) \vee \ldots \vee (a_{ik} \wedge b_{kj})$.

$$A^{[r]} = \underbrace{A \odot A \odot \ldots \odot A}_{\text{r times}}.$$

Chapter 3: Algorithms

3.2 The growth of functions

Big-O Notation $f(x)$ is $O(g(x))$ $|f(x)| \leq C|g(x)|$ $x > k$

Big-O Estimates for Important Functions

Big-Omega and Big-Theta Notation

$f(x)$ is $\Omega(g(x))$ $|f(x)| \geq C|g(x)|$

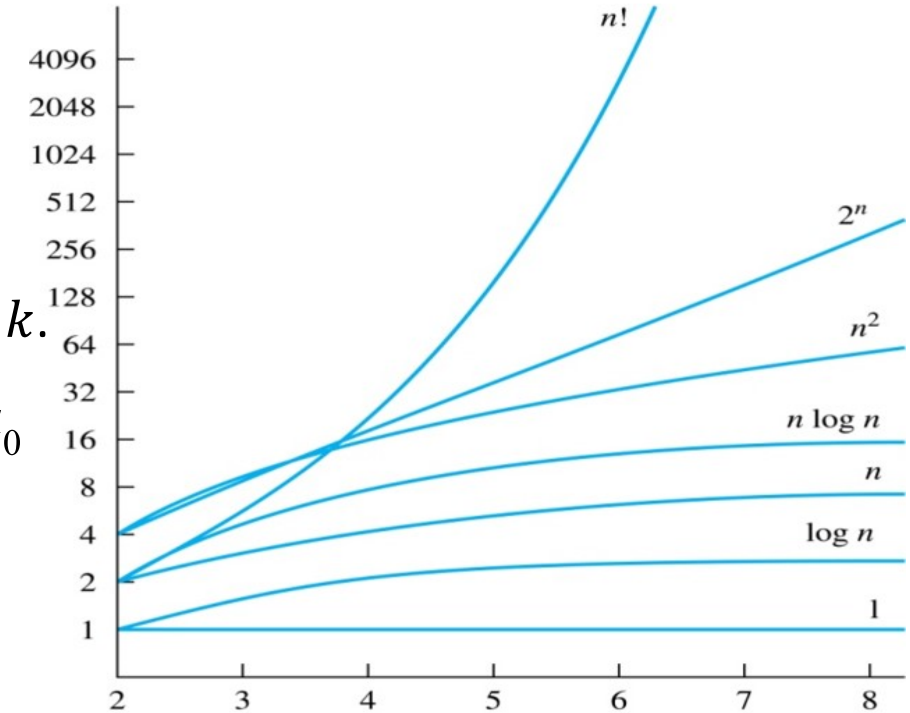
$f(x)$ is $\Theta(g(x))$

$C_1|g(x)| \leq |f(x)| \leq C_2|g(x)|$ where $x > k$.

$$f(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$$

$f(x)$ is $O(x^n)$.

$f(x)$ is of order x^n (or $\Theta(x^n)$).



3.3 Complexity of algorithms

Worst-Case Complexity

Number Theory and Cryptography(密码学)

Chapter 4

4.1 Divisibility and Modular Arithmetic

1, Division $a \mid b$

2, Division Algorithm $a = dq + r$

- Definitions of Functions **div** and **mod**

  $q = a \textbf{ div } d$

  $r = a \textbf{ mod } d$

3, Modular Arithmetic

If a and b are integers and m is a positive integer, then a *is congruent* to *b modulo m(a模m同余b)* if m divides $a - b$. $a \equiv b \pmod{m}$.

Let m be a positive integer. The integers a and b are congruent modulo m if and only if there is an integer k such that $a = b + km$.

4.2 Integer Representations and Algorithms

Integer Representations

- Base *b* Expansions
- Binary Expansions
- Octal Expansions
- Hexadecimal Expansions

Base Conversion

4.3 Primes and Greatest Common Divisors

1, Prime Numbers

2, Greatest Common Divisors and Least Common Multiples

gcd(a,b)   lcm(a,b)

3, The Euclidian Algorithm

Lemma 1: Let $a = bq + r$, where a, b, q, and r are integers. Then gcd(a,b) = gcd(b,r).

Cryptography

Section 4.6

Caesar Cipher 1

Julius Caesar created secret messages by shifting each letter three letters forward in the alphabet (sending the last three letters to the first three letters.) For example, the letter B is replaced by E and the letter X is replaced by A. This process of making a message secret is an example of *encryption*.

Here is how the encryption process works:

- Replace each letter by an integer from $\mathbf{Z}_{26}$, that is an integer from 0 to 25 representing one less than its position in the alphabet.
- The encryption function is $f(p) = (p + 3)$ **mod** 26. It replaces each integer p in the set $\{0,1,2,\ldots,25\}$ by $f(p)$ in the set $\{0,1,2,\ldots,25\}$.
- Replace each integer p by the letter with the position $p + 1$ in the alphabet.

Example: Encrypt the message "MEET YOU IN THE PARK" using the Caesar cipher.

Solution: 12 4 4 19 24 14 20 8 13 19 7 4 15 0 17 10.

Now replace each of these numbers p by $f(p) = (p + 3)$ **mod** 26.

15 7 7 22 1 17 23 11 16 22 10 7 18 3 20 13.

Translating the numbers back to letters produces the encrypted message

"PHHW BRX LQ WKH SDUN."

Caesar Cipher [2]

To recover the original message, use $f^{-1}(p) = (p-3)$ **mod** 26. So, each letter in the coded message is shifted back three letters in the alphabet, with the first three letters sent to the last three letters. This process of recovering the original message from the encrypted message is called *decryption*.

The Caesar cipher is one of a family of ciphers called *shift ciphers.* Letters can be shifted by an integer *k,* with 3 being just one possibility. The encryption function is

$$f(p) = (p + k) \textbf{ mod } 26$$

and the decryption function is

$$f^{-1}(p) = (p-k) \textbf{ mod } 26$$

The integer k is called a *key*.

Induction and recursion
（归纳和递归）

Chapter 5

5.1 Mathematical Induction

Examples of Proof by Mathematical Induction

Proving Summation Formula

Proving Inequalities2

Proving Divisibility Results

Strong Induction and Well-Ordering

Section 5.2

Strong Induction

Strong Induction: To prove that $P(n)$ is true for all positive integers n, where $P(n)$ is a propositional function, complete two steps:

- *Basis Step*: Verify that the proposition $P(1)$ is true.
- *Inductive Step*: Show the conditional statement

  $$\left[P(1) \wedge P(2) \wedge \cdots \wedge P(k)\right] \rightarrow P(k+1)$$

  holds for all positive integers k.

Strong Induction is sometimes called the *second principle of mathematical induction* or *complete induction*.

Counting(计数)

Chapter 6

6.1 The Basics of Counting

The Product Rule: A procedure can be broken down into a sequence of two tasks. There are n_1 ways to do the first task and n_2 ways to do the second task. Then there are $n_1 \cdot n_2$ ways to do the procedure.

The Sum Rule: If a task can be done either in one of n_1 ways or in one of n_2, where none of the set of n_1 ways is the same as any of the n_2 ways, then there are $n_1 + n_2$ ways to do the task.

The Subtraction Rule: If a task can be done either in one of n_1 ways or in one of n_2 ways, then the total number of ways to do the task is $n_1 + n_2$ minus the number of ways to do the task that are common to the two different ways.

the *principle of inclusion-exclusion* $|A \cup B| = |A| + |B| - |A \cap B|$

6.2 **The Pigeonhole Principle**

The Pigeonhole Principle:

If k is a positive integer and $k + 1$ or more objects are placed into k boxes, then at least one box contains two or more objects.

The Generalized Pigeonhole Principle:

If N objects are placed into k boxes, then there is at least one box containing at least $\lceil N/k \rceil$ objects.

6.3 Permutations and combinations

Permutations

The number of *r*-permutations of a set with *n* elements is denoted by $P(n,r)$.

$P(n, r) = n(n - 1)(n - 2) \cdots (n - r + 1)$

r-permutations of a set with n distinct elements.

Combinations

The number of *r*-combinations of a set with n distinct elements is denoted by $C(n, r)$.

$$C(n,r) = \frac{n!}{(n-r)!r!}.$$

Discrete Probability

Chapter 7

- With Question/Answer Animations

7.1 An Introduction to Discrete Probability

Finite Probability: If S is a finite sample space of equally likely outcomes, and E is an event, that is, a subset of S, then the *probability of E* is $p(E) = |E|/|S|$.

Probabilities of Complements and Unions of Events

$$p\left(\overline{E}\right) = 1 - p(E).$$

$$p(E_1 \cup E_2) = p(E_1) + p(E_2) - p(E_1 \cap E_2)$$

7.4 Expected Value and Variance

1,Expected Value $E(X)=\sum_{s\in S} p(s)X(s)$ $p(X=r)=\sum_{s\in S, X(s)=r} p(s)$, then

$$E(X)=\sum_{r\in X(S)} p(X=r)r.$$

2,Linearity of Expectations

$(i)\ E(X_1+X_2+\ldots+X_n)=E(X_1)+E(X_2)+\ldots+E(X_n)$

$(ii)\ E(aX+b)=aE(X)+b.$

3, Independent Random Variables

$p(X=r_1 \text{ and } Y=r_2)=p(X=r_1)\cdot p(Y=r_2),$

$E(XY)=E(X)E(Y).$

4, Variance $V(X)=\sum_{s\in S}(X(s)-E(X))^2 p(s).$ $V(X)=E((X-\mu)^2).$

$V(X)=E(X^2)-E(X)^2.$

$V(X+Y)=V(X)+V(Y)$ （X and Y are two independent random variables）